

Exercise 8.4.

1. Proof of $\angle AP = \angle PB$

In $\triangle APC$, the sum of angles is 180° :

$$\angle APC = \angle CAP + \angle ACP + \angle APC = 180^\circ$$

$$\angle BPD = \angle PDB + \angle PBD + \angle BPD = 180^\circ$$

AB and CD intersect at P .

$\angle APC = \angle BPD$ (vertically opposite)

$$\angle CAP = \angle PDB \text{ (Given)}$$

$$\angle CAP = \angle PDB$$

2. $\angle BAC = \angle ADC$

Proof of $\therefore \angle BAC = \angle DAC$ (Given AC bisects $\angle BAD$)

1) $\angle BAC = \angle DAC$ (Given: AC bisects $\angle BAD$)

2) AC (Common)

By the A.S.A congruency condition, $\triangle ABC \cong \triangle ADC$

$\angle BAC = \angle DAC$ (Corresponding)

3. Value of $\angle QOR + \angle QRO$:

$AB \parallel CD$, the interior angles on the same side of the transversal sum to 180° : $\angle BQR + \angle DRQ = 180^\circ$

The lines QO and RO are bisectors, so

$$\angle OQR = \frac{1}{2} \angle BQR \text{ and } \angle QRO = \frac{1}{2} \angle DRQ$$

$$\angle OQR + \angle QRO = \frac{1}{2} (\angle BQR + \angle DRQ) = \frac{1}{2} (180^\circ) = 90^\circ$$

In $\triangle QOR$ the sum of angles is 180°

$$\angle OQR + \angle OQR + \angle QRO = 180^\circ$$

$$\angle OQR + 90 = 180 \Rightarrow \angle OQR = 90$$

$\triangle QOR$ is a right angle triangle

4.

i) $\hat{BAE}$: In right angled ΔABC , $\hat{BAC} + \hat{ACB} = 90$. Given $\hat{ACB} = 25$ then $\hat{BAE} = 90 - 25$.

ii) $\hat{BDC} + \hat{DBC}$: In ΔBCD , the exterior angle $\hat{ADB}$ is not marked, but using ΔBCD interior sum: $\hat{BDC} + \hat{DBC} = 180 - 90 = 90$.

iii) show $\hat{BDC} + \hat{DBC} = 2\hat{BAE}$: following the geometric properties of the figure provided.

5. Triangle ABC with $\hat{A} = \hat{B} = \hat{C}$.

i) Magnitude of $\hat{BAC}$: Since the sum of angles is 180° and all three are equal, $3 \times \hat{BAC} = 180^\circ$. Thus, $\hat{BAC} = 60$.

ii) Prove AD is right angle: AD bisects $\hat{BAC}$, so $\hat{BAD} = 30$. In ΔABD , $\hat{ABD} = 60$ and $\hat{BAD} = 30$. The third angle $\hat{ADB} = 180 - (60 + 30) = 90$. Thus, ΔABD is right angled.

Miscellaneous Exercises

1.

1. $\hat{A} + \hat{B} = 110$

2. $\hat{B} + \hat{C} = 120$

3. $\hat{A} + \hat{B} + \hat{C} = 180$

From (3) and (1)

$$110 + \hat{C} = 180 \Rightarrow \hat{C} = 70$$

From (2): $\hat{B} + 70 = 120 \Rightarrow \hat{B} = 50$

From (1): $\hat{A} + 50 = 110 \Rightarrow \hat{A} = 60$

2. In ΔABC , $\hat{ABC} + \hat{ACB} = 180 - 60 = 120$

In ΔBOC , $\hat{BOC} = 180 - (\hat{OBC} + \hat{OCB})$

Since BO and CO are bisectors.

$$\hat{OBC} + \hat{OCB} = \frac{1}{2} (\hat{ABC} + \hat{ACB}) = \frac{1}{2} (120^\circ) = 60.$$

$$\hat{BOC} = 180 - 60 = 120.$$

3 Proof of $\hat{BAC} + \hat{ACB} = 2\hat{APB}$.

- ΔABC , the exterior angle at A (if BA is extended) or the relationship between interior angles is used. However, we know that in any triangle $\hat{BAC} + \hat{ACB} + \hat{ABC} = 180$

$$\overline{AP} \perp BA \text{ (Given)} \\ \hat{PAB} = 90$$

In ΔAPB , $\hat{APB} + \hat{PBA} + \hat{PAB} = 180$, since $\hat{PAB} = 90^\circ$ then $\hat{APB} + \hat{PBA} = 90$.

Since BP is the Bisector of $\hat{ABC}$.

$$\hat{PBA} = \frac{1}{2} \hat{ABC}.$$

$$\overline{AP} \hat{PBA} + \frac{1}{2} \hat{ABC} = 90 \text{ which implies } 2\hat{APB} + \hat{ABC} =$$

$\hat{BAC} + \hat{ACB} + \hat{ABC}$, we can set them equal.

$$2\hat{APB} + \hat{ABC} = \hat{BAC} + \hat{ACB} + \hat{ABC}.$$

Subtracting $\hat{ABC}$ from both sides gives the proof $2\hat{APB} = \hat{BAC} + \hat{ACB}$.

4 Proof: B Bisects $\hat{APD}$.

$$\hat{ACB} = 3x, \hat{ABC} = x, \text{ and } \hat{BAC} = 2x \text{ (Given)}$$

Sum of ΔABC :

$$2x + x + 3x = 180 \Rightarrow 2x + 4x = 180 \Rightarrow 6x = 180 \Rightarrow x = 30$$

$$AD = BD \text{ (Radii)}$$

$$\hat{ABD} = \hat{BAD} = a$$

$$a = 90^\circ - 2x$$

$$\text{Checking } \hat{DBC} = \hat{ABC}.$$

$$\hat{DBC} = \hat{ABD} - \hat{ABC} = a - x.$$

Prove the sum of interior angles of $\triangle ABC$ is 180

$PQ \parallel BC$ through point A

$\angle PAB = \angle ABC$ (Alternate angles)

$\angle QAC = \angle ACB$ (Alternate angles)

$\angle PAB + \angle BAC + \angle QAC = 180$ (Straight line PQ)

$$\angle ABC + \angle BAC + \angle ACB = 180$$